

NARRANDERA, AustraliaWMO: **957060**Lat: **34.7050S**Lon: **146.5139E**Elev: **477**StdP: **14.44**Time Zone: **10.00 (AUS)**Period: **07-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.4	32.7	24.3	18.8	69.3	26.8	21.2	61.5	23.5	59.1	20.8	56.2	0.8	140	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.1	103.6	67.2	99.1	66.5	95.4	65.6	71.9	86.8	70.4	86.0	69.2	85.4	12.0	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.8	107.6	74.8	66.5	99.3	74.7	64.4	92.1	73.9	35.9	86.8	34.6	84.4	33.6	85.8	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.2	19.6	17.5	DB	26.8	109.1	2.1	3.5	25.3	111.6	24.0	113.7	22.8	115.7	21.2	118.2	(4)
(5)			WB	26.3	74.5	2.3	2.1	24.6	76.0	23.2	77.3	21.9	78.5	20.3	80.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.7	78.9	76.7	71.5	63.0	54.4	49.2	48.1	49.5	55.0	63.2	69.8	74.1	(6)	
	DBStd	12.58	7.34	7.10	7.13	6.50	5.06	4.49	4.25	4.66	5.73	7.61	7.67	7.45	(7)	
	HDD50	247	0	0	0	1	13	68	88	66	11	0	0	0	(8)	
	HDD65	2426	1	3	23	113	329	474	523	482	306	129	34	9	(9)	
	CDD50	4890	895	749	667	392	151	44	31	49	161	411	593	747	(10)	
	CDD65	1592	431	332	225	53	1	0	0	0	7	75	178	290	(11)	
	CDH74	20651	5866	4017	2462	588	17	0	0	6	150	1173	2403	3969	(12)	
	CDH80	10566	3433	2128	1112	152	0	0	0	0	35	462	1167	2077	(13)	
(14) Wind	WSAvg	7.6	9.1	8.8	8.0	6.3	6.1	5.8	6.7	6.9	7.7	8.3	8.9	9.0	(14)	
(15) Precipitation	PrecAvg	16.5	1.4	1.3	1.1	1.4	1.7	1.5	1.5	1.6	1.5	1.7	1.2	1.3	(15)	
	PrecMax	30.6	6.1	4.7	6.1	6.7	6.0	4.6	3.3	3.3	3.9	5.2	7.2	4.0	(16)	
	PrecMin	6.7	0.1	0.0	0.0	0.0	0.2	0.1	0.1	0.0	0.4	0.0	0.0	0.0	(17)	
	PrecStd	5.3	1.3	1.4	1.2	1.4	1.3	0.9	0.9	0.9	0.9	1.2	1.3	1.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	108.5	107.7	100.0	89.1	76.7	66.2	66.6	72.0	84.9	93.7	101.9	105.7	(19)	
		MCWB	68.5	68.8	67.0	62.7	59.8	55.1	54.1	55.4	58.3	61.4	66.6	66.2	(20)	
	2%	DB	104.4	101.2	94.0	84.2	72.8	63.2	62.6	67.9	77.9	89.6	95.6	99.7	(21)	
		MCWB	67.7	66.9	65.6	61.7	57.8	54.1	53.1	53.7	56.6	60.2	64.0	66.1	(22)	
	5%	DB	100.0	96.1	90.0	80.7	69.4	60.7	59.8	63.7	73.5	85.5	91.6	95.3	(23)	
		MCWB	66.9	67.0	65.0	60.8	56.6	52.8	52.2	52.1	55.5	60.4	63.2	64.9	(24)	
	10%	DB	95.6	91.5	86.3	77.0	65.9	58.4	57.4	60.5	69.1	80.8	86.8	91.1	(25)	
		MCWB	66.2	65.8	63.6	59.9	55.3	52.2	51.1	50.9	54.0	58.4	61.9	64.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.5	74.9	71.1	67.1	62.7	57.6	58.7	58.1	61.0	66.0	70.1	72.7	(27)	
		MCDB	87.2	83.8	83.4	80.3	67.0	60.7	61.1	66.3	74.5	82.7	89.6	83.6	(28)	
	2%	WB	71.5	72.4	69.5	64.7	59.5	55.8	55.5	55.7	58.9	63.9	68.0	69.8	(29)	
		MCDB	88.1	86.2	80.6	75.1	69.4	60.1	59.6	64.9	71.1	80.4	84.4	84.1	(30)	
	5%	WB	70.1	70.6	67.5	63.0	57.6	54.3	53.3	53.4	56.9	61.9	66.3	68.2	(31)	
		MCDB	87.8	85.5	82.0	73.7	66.7	58.9	57.9	61.8	69.0	78.0	83.2	83.9	(32)	
	10%	WB	69.0	69.1	66.2	61.4	55.9	53.0	51.3	51.7	55.1	60.3	64.6	66.5	(33)	
		MCDB	87.9	84.5	81.5	74.0	64.2	57.5	56.7	59.4	67.2	77.3	79.5	84.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	27.1	25.0	24.4	24.6	22.2	19.0	18.6	21.4	25.9	28.0	26.7	26.8	(35)	
		MCDBR	32.0	30.7	28.6	28.3	25.6	22.1	22.0	26.6	32.2	33.6	32.5	31.5	(36)	
	5% WB	MCWBR	9.8	9.7	10.2	12.0	13.1	13.8	13.5	14.6	14.9	12.9	10.4	10.3	(37)	
		MCWBR	24.2	21.0	21.9	22.4	21.7	17.5	17.4	23.1	25.9	27.0	24.5	22.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.345	0.342	0.321	0.309	0.297	0.289	0.286	0.293	0.323	0.327	0.344	0.338	(40)	
		taud	2.478	2.517	2.558	2.556	2.560	2.589	2.596	2.555	2.452	2.472	2.462	2.489	(41)	
	Ebn at Noon		314	309	304	290	276	270	278	291	298	310	312	318	(42)	
		Edh at Noon	37	34	30	27	23	22	23	26	33	35	37	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		2472	2175	1837	1359	981	757	830	1130	1582	2007	2267	2494	(44)	
	RadStd		141	121	102	60	49	59	50	87	121	137	152	90	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		+0.83	N/A	N/A	+2.09	N/A	N/A	N/A	N/A	+243	N/A

Nomenclature: See separate page